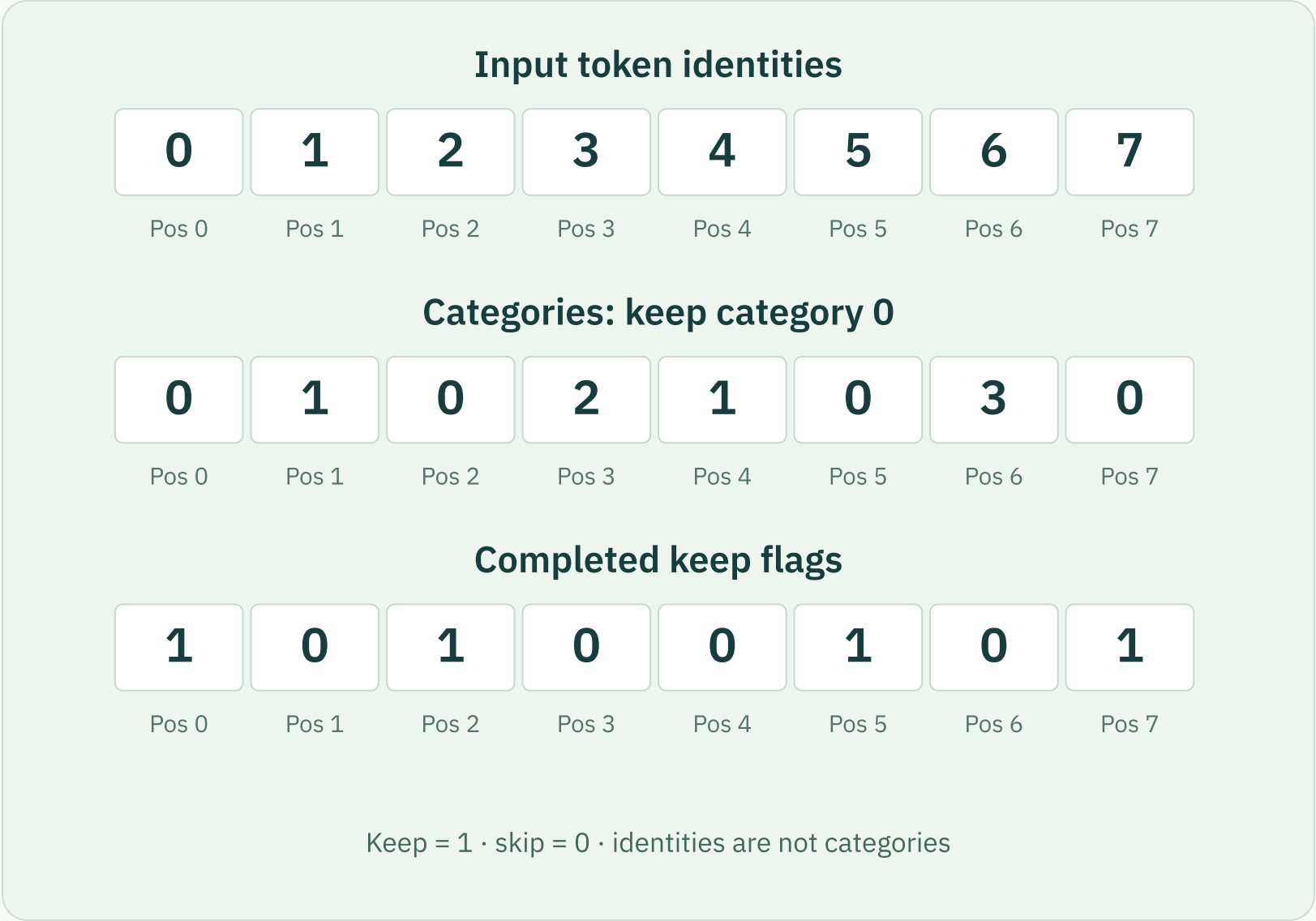
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Keep identities, not just a count.

Compaction packs selected items into consecutive output slots.

> Completed flags and offsets before scatter · unique destinations · no GPU timing



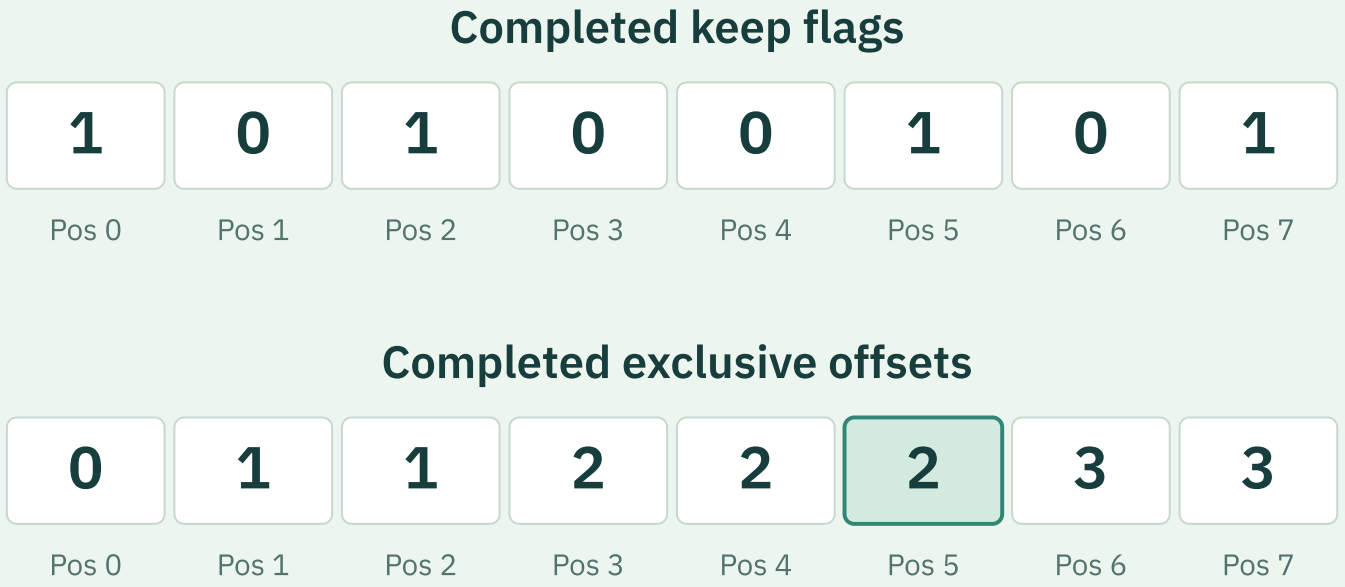
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Count the earlier survivors.

An exclusive scan allocates a destination for each kept token.

> Completed flags and offsets before scatter · unique destinations · no GPU timing



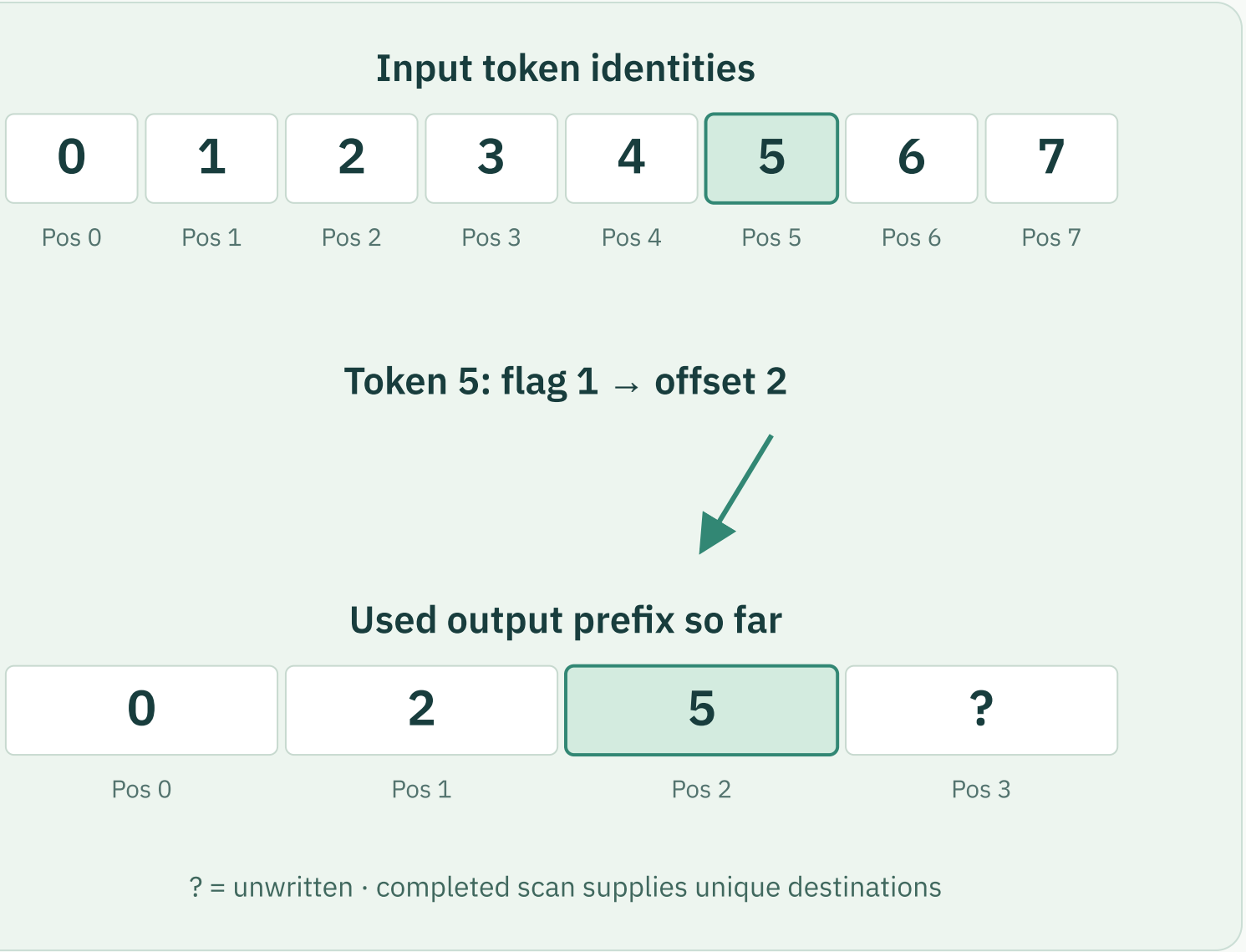
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Write one identity to its own slot.

Token 5 uses offset 2 as its output destination.

> Completed flags and offsets before scatter · unique destinations · no GPU timing



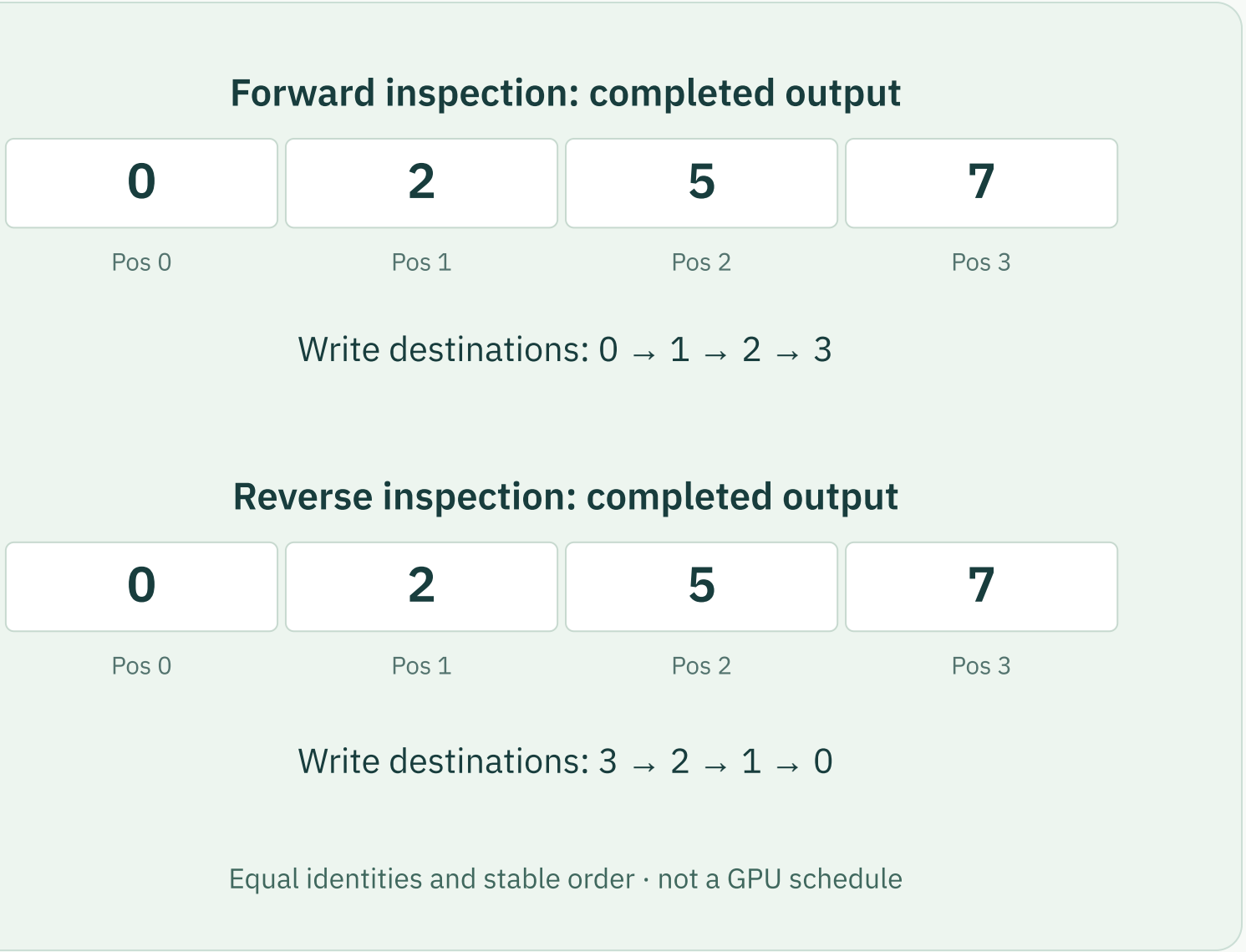
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Write order need not be input order.

Fixed destinations preserve stable output under reversed inspection.

> Completed flags and offsets before scatter · unique destinations · no GPU timing



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Pause: change only the selection.

Keep every category except zero; retain reversed write inspection.

> Completed flags and offsets before scatter · unique destinations · no GPU timing

Input categories: keep category ≠ 0

Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7
0	1	0	2	1	0	3	0

Keep flags

Pos 0	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7
0	1	0	1	1	0	1	0

Output identities: ?

Token 4 destination: ?

Reverse write inspection · predict before comparison

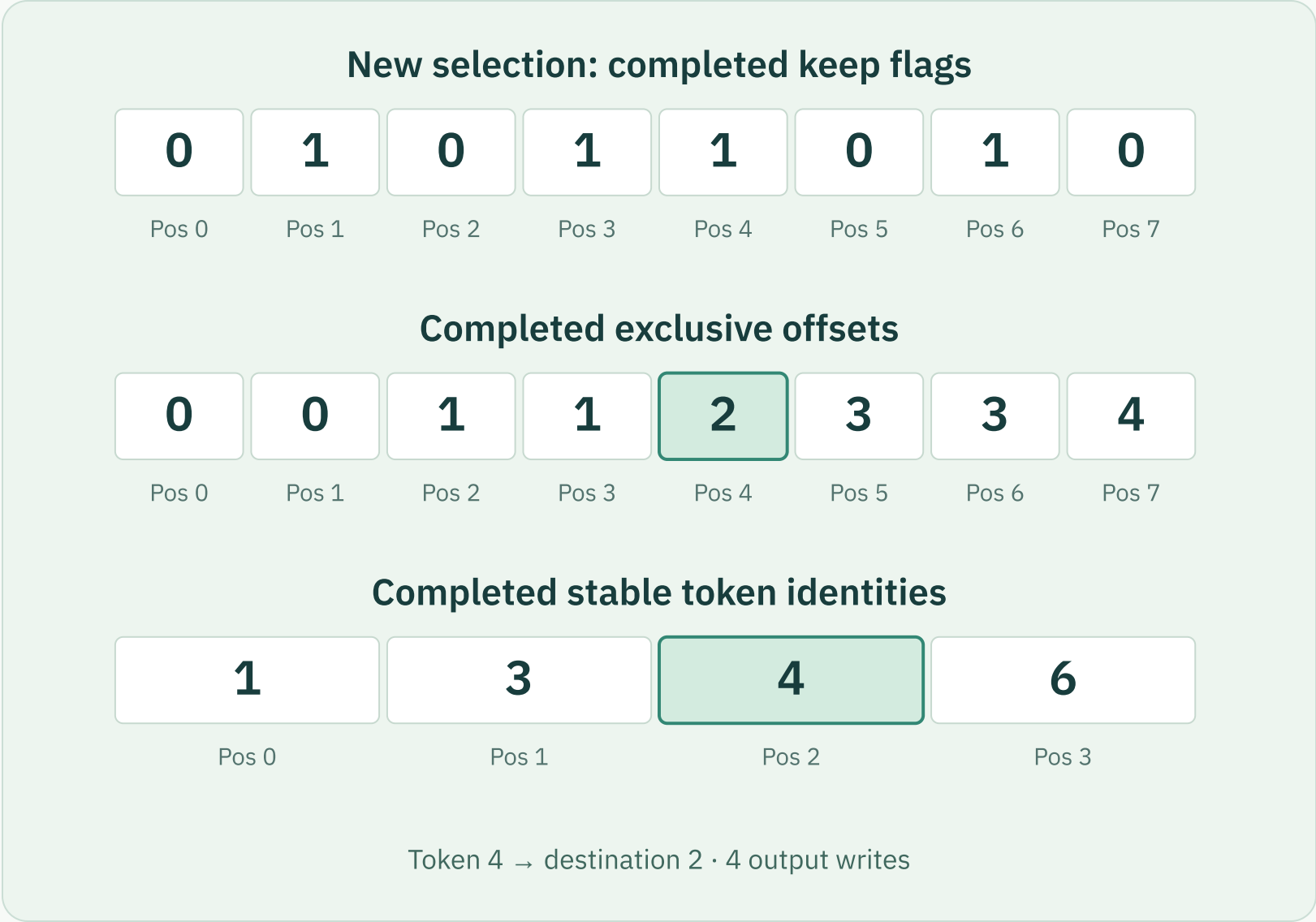
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Offsets preserve the new survivor order.

Token 4 writes slot 2; skipped tokens make no writes.

> Completed flags and offsets before scatter · unique destinations · no GPU timing



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